

1 Summary

Model (Run-name): model1 (test)

Iteration: 13

TAC: 154688.7415551182

Took: 1.9902198314666748s

Total Time: 17.02483057975769s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

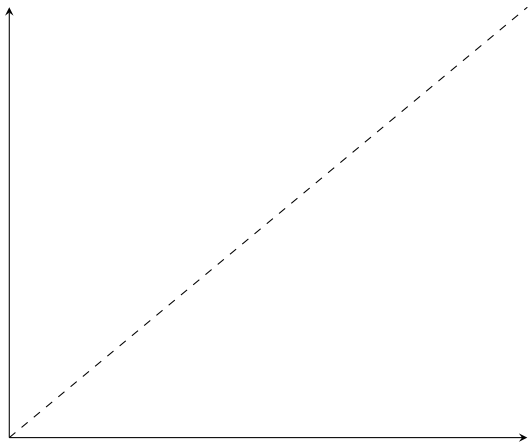
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

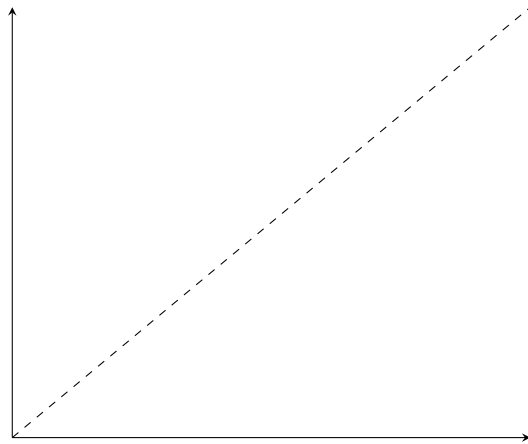
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05249040946317568	$b_{1,1,1}^{H,out}$	887.6021908041035
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.01741370394782146	$b_{1,2,2}^{H,out}$	3878.3010447968945
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.02984213557668777	$b_{2,1,2}^{H,out}$	9320.907100992757
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014511551676778922	$b_{1,1,1}^{C,in}$	5976.723689957428
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.00960338006077692	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.02260591884550999	$b_{2,1,2}^{C,in}$	4215.076263196301
$A_{1,1,1}^\beta$	70.22290217272871	$q_{1,1,1}$	671.6989552031054	$b_{1,1,1}^{C,out}$	6648.422645160533
$A_{1,2,2}^\beta$	67.91344539650363	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	144.70843093515643	$q_{2,1,2}$	2442.4515671953422	$b_{2,1,2}^{C,out}$	6657.527830391643
$A_{cu,1}^\beta$	4.993068227518925	$q_{cu,1}$	178.30104479689453		
$A_{cu,2}^\beta$	37.554741031910254	$q_{cu,2}$	1957.5484328046578		
$A_{hu,1}^\beta$	13.161020122653593	$q_{hu,1}$	485.8494776015523		
$A_{1,1,1}$	70.22290217272871	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	67.91344539650363	$t_{1,2}^{(H)}$	582.8301044796895		
$A_{2,1,2}$	144.70843093515643	$t_{1,3}^{(H)}$	387.83010447968945		
$A_{cu,1}$	4.993068227518925	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.554741031910254	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.161020122653593	$t_{2,3}^{(H)}$	467.8774216402329		
$dt_{1,1,1}$	32.38996517343682	$t_{1,1}^{(C)}$	617.6100348265632		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	572.8301044796895		
$dt_{1,2,2}$	82.83010447968945	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	37.83010447968945	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	17.169895520310547	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	57.87742164023288	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	67.83010447968945	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	147.87742164023288	$t_{1,2,2}^H$	387.83010447968945		
$dt_{hu,1}$	62.38996517343682	$t_{2,1,2}^H$	467.4953938558227		
$f_{1,1,1}^H$	2.3989248400110905	$t_{1,1,1}^C$	636.9614739994557		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	19.93789604777644	$t_{2,1,2}^C$	645.9307590714515		
$f_{1,1,1}^C$	10.437715492297125	$b_{1,1,1}^{H,in}$	1559.301146007209		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5828.3010447968945		
$f_{2,1,2}^C$	10.280673812673903	$b_{2,1,2}^{H,in}$	11763.3586681881		

3 *RecLMTD* Tangent Points

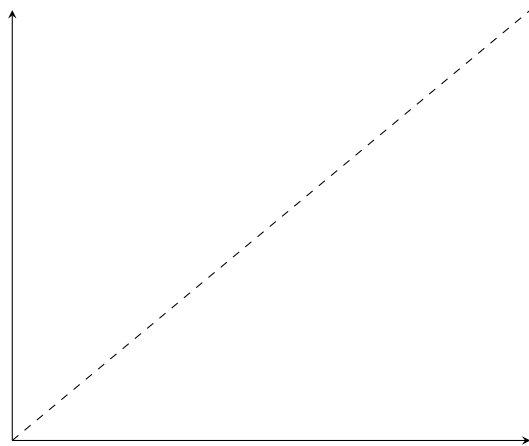
- $(1, 1, 1)$
 - Value: 0.052490
 - Error: 0.000000228903567
 - Relative error: 0.000004360845550
- $(1, 2, 2)$
 - Value: 0.017414
 - Error: 0.000001548841561
 - Relative error: 0.000088935921838
- $(2, 1, 2)$
 - Value: 0.029842
 - Error: 0.000009095389058
 - Relative error: 0.000304690586061
- $cu, 1$
 - Value: 0.014512
 - Error: 0.000000265358189
 - Relative error: 0.000018285662547
- $cu, 2$
 - Value: 0.009603
 - Error: 0.000000025401661
 - Relative error: 0.000002645068081
- $hu, 1$
 - Value: 0.022606
 - Error: 0.000000066135709
 - Relative error: 0.000002925584058



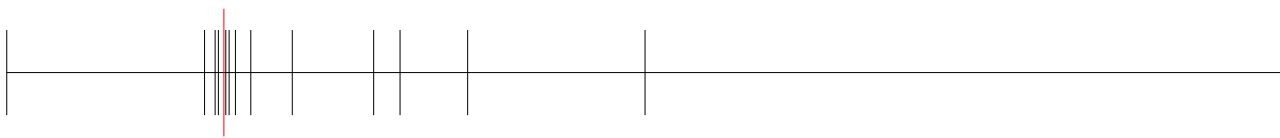
(a) *RecLMTD* - (1, 1, 1)



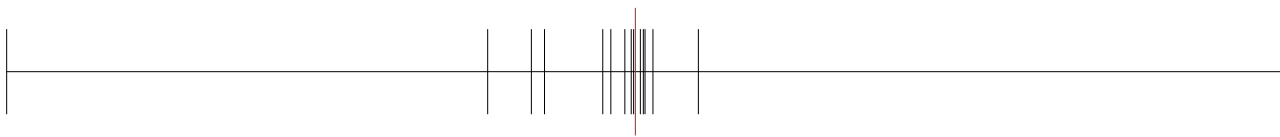
(b) *RecLMTD* - (1, 2, 2)



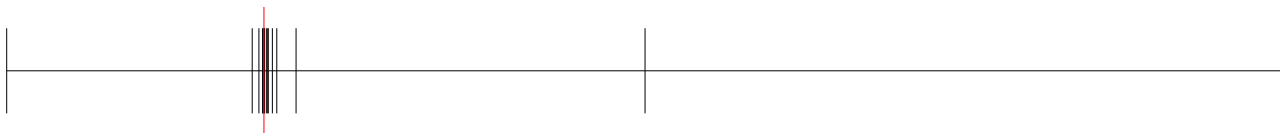
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1559.301146007209
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5828.3010447968945
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 11763.3586681881
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 887.6021908041035
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3878.3010447968945
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9320.907100992757
 - Absolute error: 0.03253548105931259
 - Relative error: 3.4906038945849123e-06

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 5976.723689957428
 - Absolute error: 2.313966024406909
 - Relative error: -0.00038701312109848656
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4215.076263196301
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 6648.422645160533
 - Absolute error: 1.8189894035458565e-12
 - Relative error: 2.7359713734052715e-16
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6657.527830391643
 - Absolute error: 16.9243908051958
 - Relative error: 0.0025486224195085784



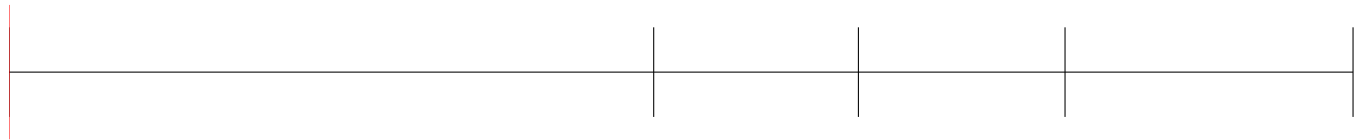
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



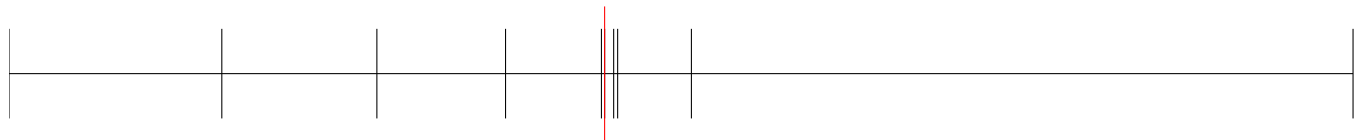
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



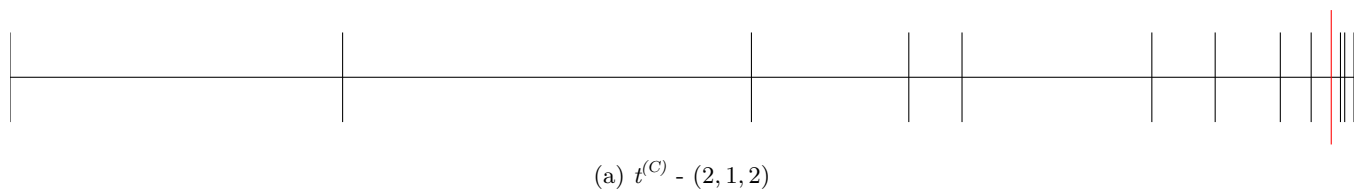
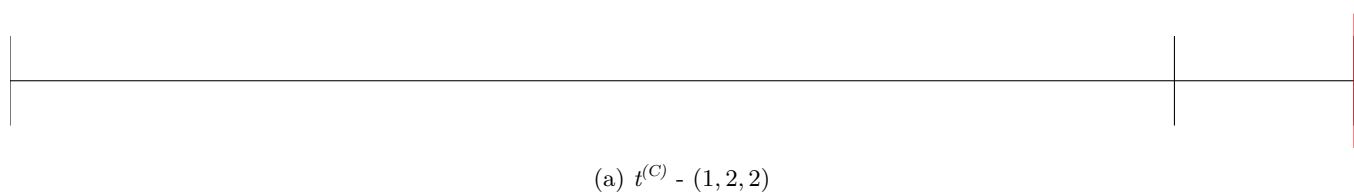
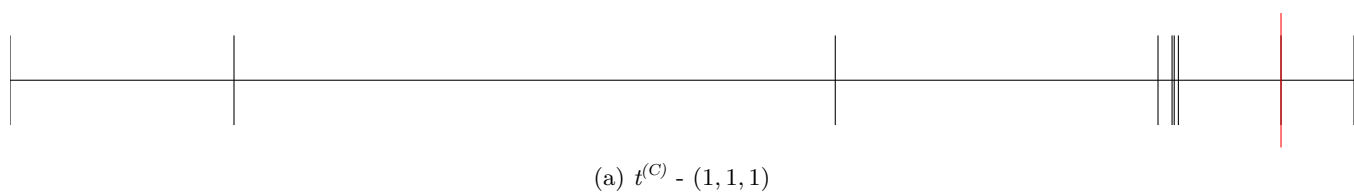
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

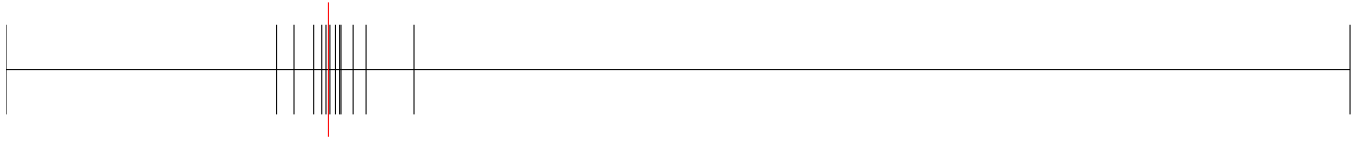
- $(1, 1, 1)$
 - Value: 70.222902
 - Error: 0.292604216467893
 - Relative error: 0.004149501740127
- $(1, 2, 2)$
 - Value: 67.913445
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 144.708431
 - Error: 1.067510680317412
 - Relative error: 0.007322955135720

5.2 A_{CU}

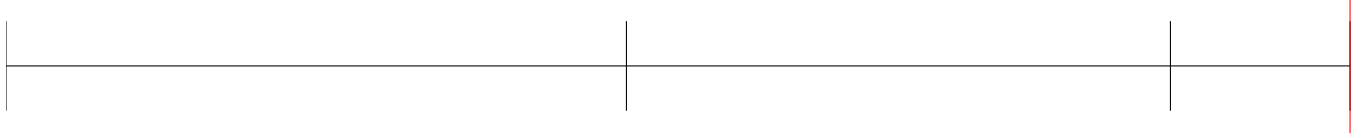
- $cu, 1$
 - Value: 4.993068
 - Error: 0.181781423668692
 - Relative error: 0.035127865720113
- $cu, 2$
 - Value: 37.554741
 - Error: 0.043422143292460
 - Relative error: 0.001154900655389

5.3 A_{HU}

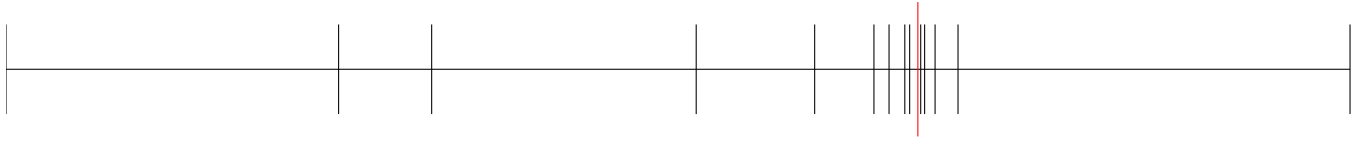
- $hu, 1$
 - Value: 13.161020
 - Error: 0.018668511499344
 - Relative error: 0.001416460738759



(a) $A^\beta - (1, 1, 1)$



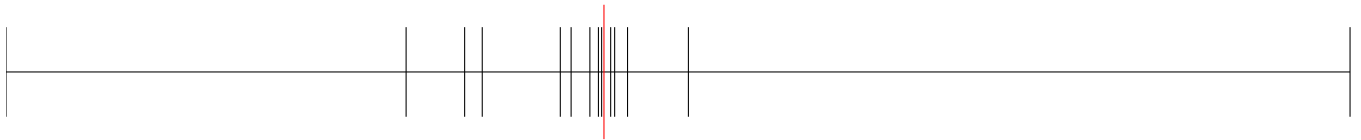
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

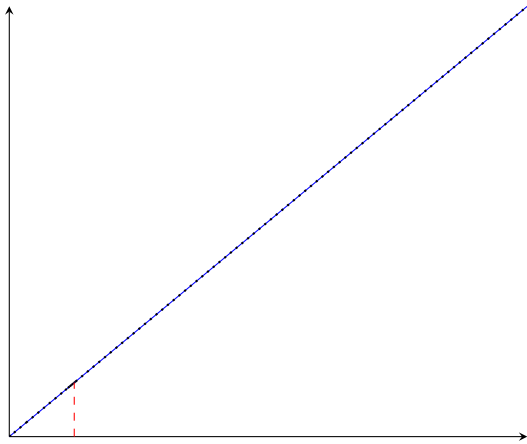


(a) $A^\beta - hu, 1$

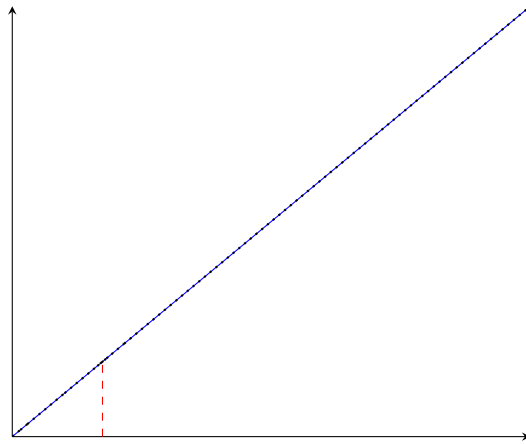
6 A^β Breakpoints

- (1, 1, 1)
 - q : 70.222902
 - $\hat{q}^\beta$: 70.222902
 - q^β : 70.222902
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, **69.89902342903967**, **70.30181758862284**, 71.22933662920877, 71.6637831223933, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 67.913445
 - $\hat{q}^\beta$: 67.913445
 - q^β : 67.913445
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, **67.27673099173637**, **69.34502840170047**, 70.94081287188642, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 144.708431
 - $\hat{q}^\beta$: 144.708431
 - q^β : 144.708431
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 131.69760457629607, **132.12228195956155**, **147.07009017600456**, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.993068
 - $\hat{q}^\beta$: 4.993068
 - q^β : 4.993068
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 4.269280092716329, 4.432280503220532, **4.983938626310074**, **5.0418124100090695**, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.554741
 - $\hat{q}^\beta$: 37.554741

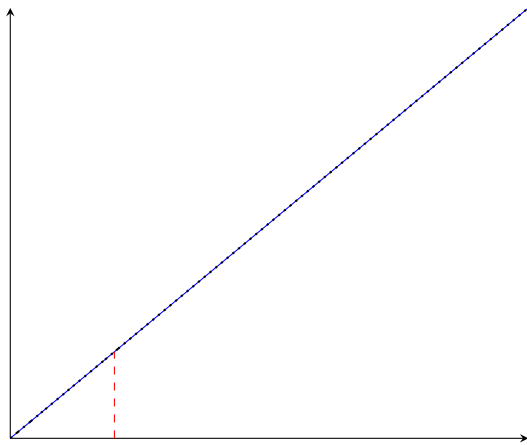
- q^β : 37.554741
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, **37.47875108132638**, **37.9454585315849**, 38.21652191748154, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.161020
 - $\hat{q}^\beta$: 13.161020
 - q^β : 13.161020
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, **13.104034300095861**, **13.192252946941991**, 13.22717920580809, 13.244985069124171, 13.30871858348944, 13.346967780023082, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



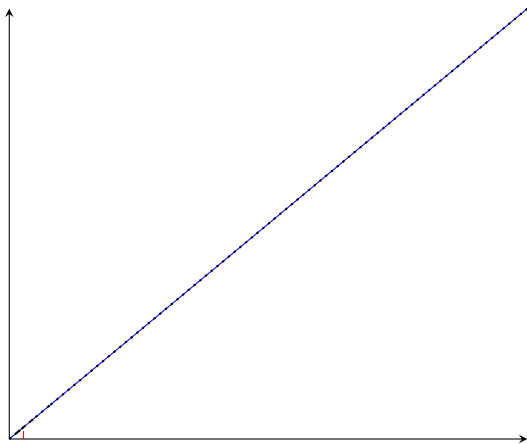
(a) $q^\beta - (1, 1, 1)$



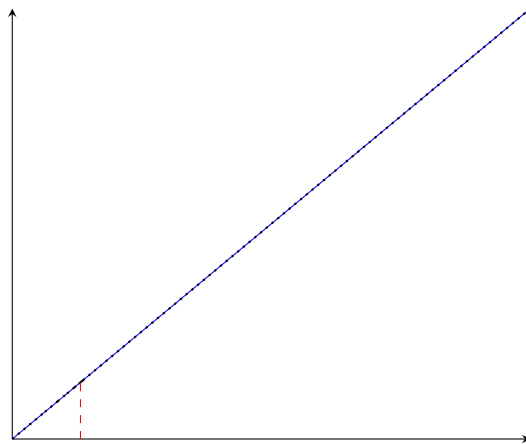
(b) $q^\beta - (1, 2, 2)$



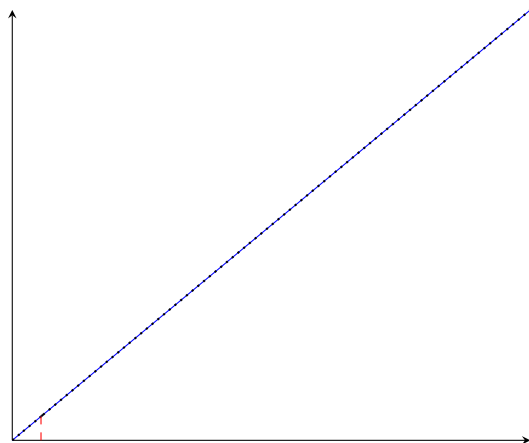
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



(a) $q^\beta - hu, 1$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	36.64133181190118
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1934.9237368036995
$z_{2,1,1}$	0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008057646108643299	$b_{2,1,1}^{C,in}$	2615.7278772379145
$z_{2,2,2}$	0.0	$RecLMTD_{2,1,1}$	0.1	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{1,1,2}^{C,out}$	1934.9237368036984
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	2615.7278772379145
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	370.0		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	523.9167304164481		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	575.1418898496516		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4940.698853992792		
$f_{1,2,1}^H$	7.601075159988909	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	36.64133181190118		
$f_{2,2,2}^H$	0.062103952223559133	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.7193261873260965	$b_{1,2,1}^{H,out}$	4940.698853992791		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	4.562284507702875	$b_{2,2,1}^{H,out}$	0.0		